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## Characterization of plasma in microwelding of glass using ultrashort laser pulse at high pulse repetition rates

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### Abstract

Dynamic behavior of plasma in internal modification of glass using fs-laser pulses with high pulse repetition rates up to 1MHz is observed using high-speed video camera at 50,000 frames per second. High-speed pictures reveal striking fact that the localized plasma is produced with size much smaller than reported in literatures before, and changes its location, size, shape and brightness periodically at repetition rates of the order of kHz. The mechanism responsible for producing localized plasma and its long-term periodic motion is presented based on the analysis of free-electron density based on rate equation.

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**Keywords:** microwelding; nonlinear absorption; rate equation; high-speed camera; free-electron density; critical electron density

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### 1. Introduction

Internal modification of transparent material using ultrashort laser pulse (USLP) has attracted much attention due to interesting applications such as microwelding [1-3], waveguide formation [4,5], selective etching [6] and so on. Among these applications, microwelding of glass using USLP is considered to be especially promising, since crack-free welding with monolithic structure having high mechanical strength is available even with glass having high

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thermal expansion coefficient. Performances of these applications are strongly dependent on the physical properties of the laser-induced plasma. Since the laser energy absorption depends on the spatial and temporal distribution of free-electron density in plasma [7,8], characterization of the plasma is one of the most important tasks for understanding and optimizing the modification process. Laser-induced plasma has been characterized mainly in single pulse irradiation or multipulse irradiation at low pulse repetition rates by simulating the evolution of free-electron density based on rate equation [9-11], and experimental measurement using pump-probe technique [12,13].

However, internal modification using USLP with high pulse repetition rates is rather complicate, where the modified region with dual-structure [14], which consists of teardrop-shaped inner structure and elliptical outer structure. The laser energy is absorbed by nonlinear process in the inner structure [14], which can be much larger than focus volume of USLP. Physical process in the modified region has been analysed by the experimental measurement of nonlinear absorptivity [15], and simulation based on thermal conduction model [16,17].

It is also known it takes some time of the order of *ms* before steady temperature distribution is established [14] with accompanying some fluctuation of laser-matter interaction [20,21]. Therefore difficulty is found to use existing pump-probe technique [9,10] for characterizing plasma at high pulse repetition rates. While free-electron density in the plasma is also simulated at high pulse repetition rate [22], process fluctuation is not taken into consideration.

In the present study, high-speed video camera is used to observe the dynamic behaviour of plasma with time scale as long as *ms* order, which has not been observed in the past, while the temporal resolution the camera is not high enough to observe quick phenomena of the time scale of MHz or higher. The effects of experimental parameters such as pulse energy, pulse repetition rate and translation speed of sample on the periodic evolution of plasma are also discussed. The mechanism of periodical plasma evolution is discussed based on the analysis of free electron density using rate equation model.

## 2. Experimental procedure

*fs*-laser pulses from IMRA (pulse duration  $\tau_p = 550$  fs, wavelength  $\lambda = 1045$  nm) is focused into borosilicate glass plate (D263, Schott) with a thickness of 1 mm using NA0.65 lens at a depth of approximately 250  $\mu$ m. The *fs*-laser pulses are irradiated to the side plane of the glass plate in perpendicular to the thickness direction. We used a high-speed video camera (Photron Fastcam) providing 50,000 frames per second and exposure time of 5.4  $\mu$ s. The filming provides pictures at each 20 pulses with overlapped image of 5 ~ 6 pulses, when the *fs*-laser pulses are irradiated at a repetition rate of 1 MHz, for instance. Although our photographing procedure is not high enough to observe quick changing processes in pulse-by-pulse basis, a series of pictures provide a new opportunity to observe dynamic behavior of plasma with longer time scale, which has not been observed in the past.

Pictures were taken through band pass filter for 532 nm with and without irradiating SHG Nd:YAG laser. When SHG Nd:YAG laser is irradiated, some attenuation of SHG Nd:YAG laser in the plasma region is observed suggesting the plasma is of high density free-electrons [21,22]. However, the quality of the pictures is not good enough to evaluate the absorption property from the pictures, because of fringe patterns appeared in the pictures. In addition, the light emission from the hot plasma can compete with the SHG Nd:YAG laser beam transmitted through the band pass filter, since rather strong plasma emission lasts between the *fs*-laser pulses due to the heat accumulation effect at high pulse repetition rates [14,21]. Thus pictures of the laser-induced plasma taken without SHG Nd:YAG laser are solely shown in the paper, and the evolution of free-electron density in conduction band is estimated by the simulation based on the rate equation model [10,11,22].

## 3. Results and discussions

### 3.1. Cyclic behavior of plasma evolution

**Figures 1(a)** and **(b)** show typical cross-section and side view of the internally modified glass sample obtained at pulse repetition rate  $f = 1$  MHz, pulse energy  $Q = 2.56$   $\mu$ J and translation speed  $v = 20$  mm/s. The cross-section shows a dual-structure [14,16] consisting of a teardrop-shaped inner structure and an elliptical outer structure. Thermal conduction model [2,14,16] indicates that the laser energy is absorbed in the inner structure, and that the glass is melted in the outer structure, where working temperature with a viscosity of  $10^4$  dPas is reached. The dual-

structure is also clearly observed in the side view (Fig. 1b) where the vertical size of the inner structure changes at some distance after the beginning of the laser irradiation, and it takes several thousands of the laser pulses for the vertical size to reach steady value.

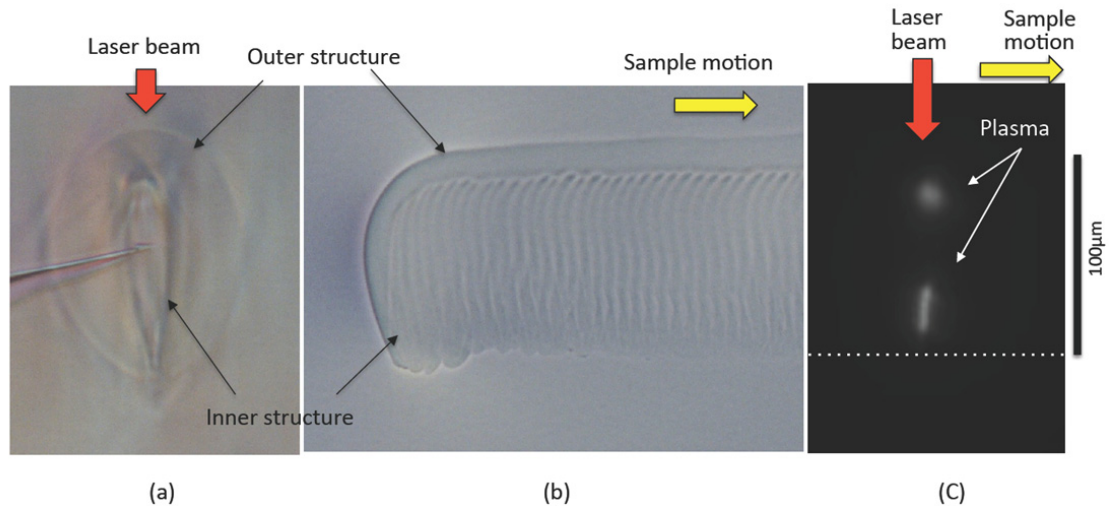


Fig. 1. (a) Cross-section of internal modification. The sample moves perpendicularly to the paper plane; (b) Side-view of the modified glass sample. Vertical sizes of inner and outer structure vary at the beginning of approximately  $100\ \mu\text{m}$ . (c) A snapshot of laser-induced plasma taken from side. The location of geometrical focus is shown by a dotted line. ( $f=1\text{MHz}$ ,  $Q=2.56\ \mu\text{J}$ ,  $v=20\text{mm/s}$ ).

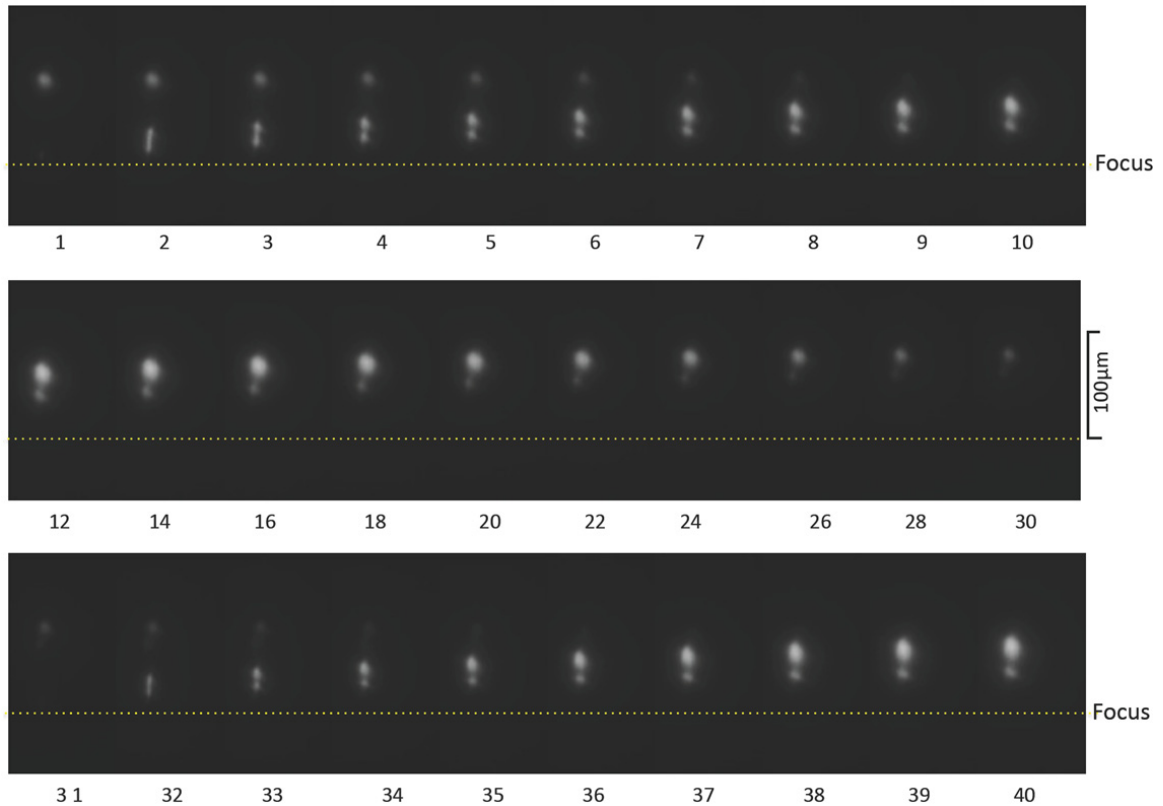


Fig. 2. High-speed pictures taken at a frame rate of 50,000 per second under conditions of  $f=1\text{MHz}$ ,  $Q=2.56\text{ }\mu\text{J}$ ,  $v=20\text{ mm/s}$  in D263. Figures indicate the relative number of the frame.

**Figure 1(c)** shows a snap shot where interestingly two separate plasma images are observed. The picture shows the plasma appears within the region of the internal structure as predicted in Ref.[14,15]. However, the plasma does not fill the inner structure, but concentrates locally in a small region of the internal structure. This suggests the temperature of the plasma is much higher than the value simulated based on thermal conduction model [14,15] and rate equation [22] where the plasma is assumed to fill the inner structure of the plasma.

**Figure 2** shows a series of high-speed pictures taken under the same condition as Fig.1. The high-speed pictures reveal very striking fact that the plasma evolution in the inner structure is not uniform but is localized with cyclic change in height, dimensions, and brightness. The cyclic plasma evolution is characterized by generation of plasma near the focus (#2), upward expansion (#3~#24) and disappearance at the highest position (#26~#31). It is noted that long and thin linear plasma produced near the focus expands upward rapidly (#2 and #6) when a plasma spot in the last cycle still exists at the highest position. Then the expansion speed is slowed down gradually with changing its shape from linear to sphere; sometimes it breaks down into two smaller pieces. The plasma becomes brightest after the maximum height. Then the plasma decreases its brightness and the size, and disappears eventually at the highest position. It is interesting to note that the plasma is produced near when the less bright plasma in the previous cycle still exists and two plasmas coexist at different heights as seen in #2~#7. This suggests that the upper plasma is transparent.

**Figure 3** shows the change in the height of the plasma at its top and bottom with respect to the geometrical focus from the first laser pulse. It is noted that the frequency of the cyclic plasma evolution is two to three orders smaller than the repetition rate of the  $fs$ -laser pulses. Change of the maximum height and cycle of the plasma evolution is clearly observed. The figure also shows steady height of the inner and outer structure are reached after 3,000~4,000 pulses in 1MHz condition, in accordance with the variation of vertical size of the inner structure shown in Fig. 1(b).

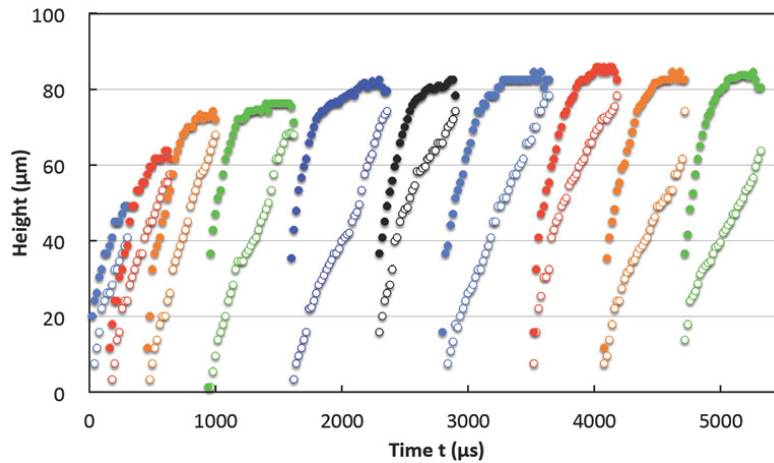


Fig. 3. Vertical position of the plasma at its top and bottom with respect to the geometrical focus is shown by closed and open circles, respectively, plotted from the first laser pulse ( $f=1\text{MHz}$ ,  $Q=2.56\text{ }\mu\text{J}$  and  $v=20\text{ mm/s}$ , D263).

**Figure 4** shows upward velocity of the upper and lower edge of the plasma determined in a typical cycle in Fig. 3, which is determined from each 3 photographs to reduce the scattering of evaluation. The plasma evolution is divided into three phases consisting of expansion phase I starting from plasma generation, equilibrium phase II and expiring phase III. In phase I, the velocity of the top edge  $v_U$  is faster than that of the bottom edge  $v_L$ , indicating the plasma is expanding. The maximum upward velocity of the plasma is as fast as approximately  $v_U \approx 400\text{ mm/s}$ , which corresponds to approximately  $0.4\text{ }\mu\text{m}$  per  $\text{fs}$ -laser pulse. In phase II  $v_U$  is nearly equal to  $v_L$ , indicating the plasma size keeps nearly constant, the brightness of the plasma increases as is seen in #11~#14 in Fig. 2. In phase III where the plasma stays near the highest position  $h_{\text{max}}$ , brightness of the plasma is gradually reducing. A new bright plasma is produced near the focus by the laser energy transmitted through the upper plasma before complete expiration. This suggests that phase III is cooling period of the plasma.

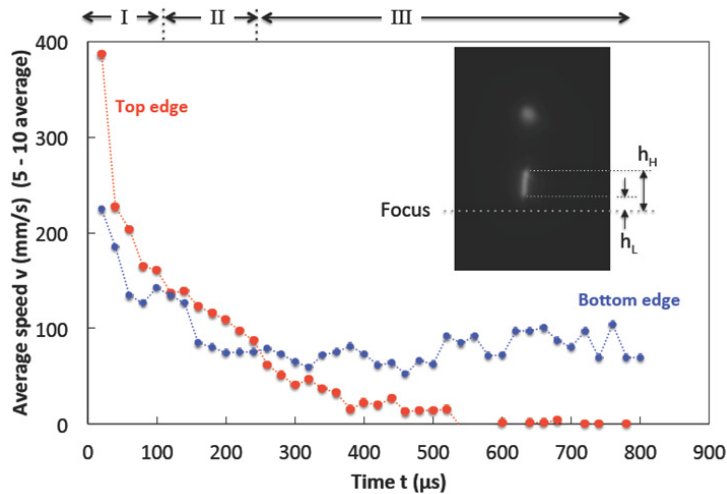


Fig. 4. Typical moving speed of the plasma at the upper and lower edges at  $f=1\text{MHz}$ ,  $Q=2.56\text{ }\mu\text{J}$  and  $v=20\text{ mm/s}$ , corresponding to Figs. 3 and 4. The plasma evolution is divided into three periods consisting of (I) glowing, (II) equilibrium and (III) extinction periods.

### 3.2. Effects of process parameters

In this section, the effect of the process parameters on the frequency of plasma evolution  $f_p$  is discussed. The process parameters include pulse energy  $Q$  and pulse repetition rate of  $fs$ -laser  $f_s$  and translation speed  $v$ . In **Fig. 5**  $f_p$  is plotted vs. pulse energy  $Q$  at  $f_s = 1\text{ MHz}$  and  $v = 20\text{ mm/s}$ . While the frequency is as large as  $f_p \approx 2\sim 2.5\text{ kHz}$  at  $Q = 1\text{ }\mu\text{J}$ , it decreases as  $Q$  increases down to  $f_p \approx 400\text{ Hz}$  at  $Q = 6.5\text{ }\mu\text{J}$ . It is seen that  $f_p$  decreases in inversely proportional to  $v$  as a dotted line is shown. The experimental results show that the vertical size of the inner structure is proportional to  $Q$ , suggesting that average moving speed of the plasma starting from the bottom to the top of the inner structure is unchanged.

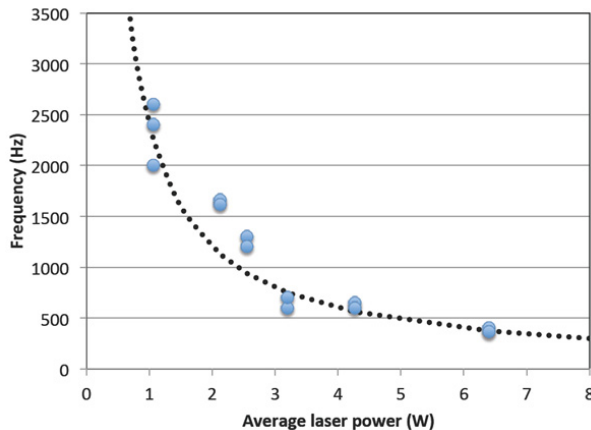


Fig. 5. Frequency of plasma evolution  $f_p$  plotted vs. pulse energy  $Q$  at  $1\text{ MHz}$  and  $20\text{ mm/s}$ . The frequency  $f_p$  changes in inversely proportion to average laser power  $\bar{W}$  shown by a dotted line.

**Figure 6** shows the effect of the repetition rate of the  $fs$ -laser pulses  $f$  on  $f_p$  at  $Q = 2.56\text{ }\mu\text{J}$  and  $v = 20\text{ mm/s}$ . The value  $f_p$  decreases as  $f$  increases. **Figure 7** shows the effect of  $v$  on  $f_p$  at  $Q = 2.56\text{ }\mu\text{J}$  and  $f_s = 1\text{ MHz}$ , showing that  $f_p$  decreases as  $v$  decreases. These two pictures indicate that  $f_p$  decreases as the size of the inner structure increases, in accordance with the tendency shown in Fig. 5.

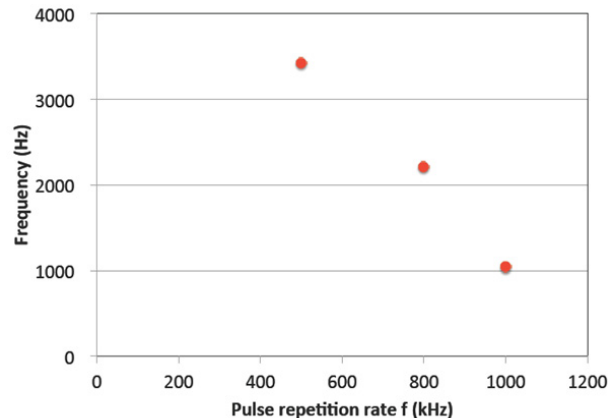


Fig. 6. Effect of pulse repetition rate of  $fs$ -laser  $f$  on the repetition rate of periodical plasma motion  $f_p$  at  $Q = 2.56\text{ }\mu\text{J}$  and  $v = 20\text{ mm/s}$ .

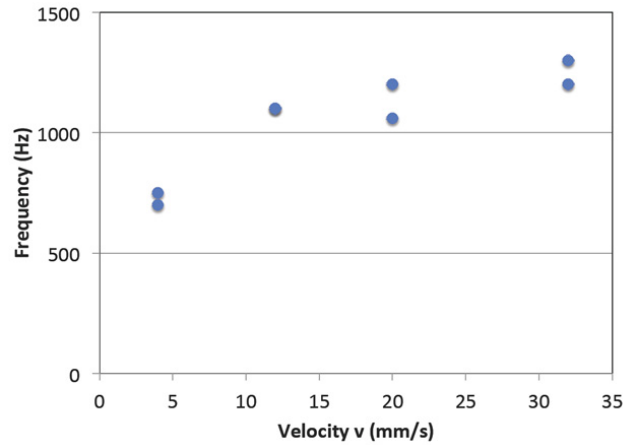


Fig. 7. Effect of translation speed  $v$  the repetition rate of periodical plasma motion  $f_p$  at  $Q=2.56\mu J$  and  $f=1MHz$ .

### 3.3. Mechanism of cyclic motion of plasma

#### (a) Basic process of upward motion

In order to understand the mechanism of the cyclic motion of the plasma shown in Fig. 2, deposition of the laser energy in plasma is analyzed. The time evolution of the free electron density  $\rho(t)$  in conduction band under the influence of the laser beam is simulated based on the rate equation given by [10,11,22]

$$\frac{\partial \rho(t)}{\partial t} = \eta_{photo} I^k(t) + \eta_{casc} I(t) \rho(t) - \eta_{rec} \rho(t)^2 \quad (1)$$

where  $\eta_{photo}$ ,  $\eta_{casc}$  and  $\eta_{rec}$  are coefficient of photoionization, cascade ionization and recombination of free-electrons, respectively,  $k$  is photon number for MPI given by  $\langle E_g / \hbar\nu + 1 \rangle$  where  $E_g$  is band gap energy and  $\langle \rangle$  indicates the integer part within it, and  $I(t)$  is laser irradiance given by

$$I(t) = I_0 \exp \left\{ -4 \ln 2 \frac{(t - \tau_p)^2}{\tau_p^2} \right\}. \quad (2)$$

$\tau_p$  is duration of the laser pulse. The cascade ionization coefficient  $\eta_{casc}$  is given by [10]

$$\eta_{casc} = \frac{1}{\omega^2 \tau^2 + 1} \left[ \frac{e^2 \tau}{c n_0 \epsilon_0 m (3/2) \tilde{\Delta}} I(t) - \frac{m \omega^2 \tau}{M} \right] \quad (3)$$

where  $\tau$  is the time between collision for inverse Bremsstrahlung,  $c$  the vacuum speed of light,  $n_0$  the refractive index of the medium at laser-frequency  $\omega$ ,  $m$  electron mass,  $M$  molecular mass,  $\epsilon_0$  vacuum permittivity, and  $\tilde{\Delta}$  the effective ionization potential.

The time-dependent free-electron density  $\rho(t)$  simulated at a location of  $z = 12.6 \mu m$  is plotted in **Fig. 8(a)**, when the single laser pulse of  $Q = 2.56 \mu J$  is focused into bulk glass at location of  $z = 0$ , assuming shielding effects of the laser-induced plasma is negligible, for simplicity, and the radius of focused laser beam spot at  $z$  is given by



$$\omega(z) = \omega_0 \sqrt{1 + \left( \frac{M^2 \lambda z}{\pi \omega_0^2 n_g} \right)^2}; \quad \omega_0 = \frac{M^2 \lambda}{\pi NA}, \quad (4)$$

where  $\omega_0$  is laser spot radius at  $z = 0$ ,  $M^2$  beam quality factor, NA numerical aperture of lens,  $\lambda$  wavelength of laser beam and  $n_g$  refractive index. In the simulation,  $\eta_{photo}$  given by Kennedy's approximation [11] of Keldysh model [23] is used. The free-electron density produced by multiphoton ionization simulated without the contribution of avalanche ionization  $\rho_{photo}$  is also plotted in the figure. It is seen that  $\rho_{max}$ , which is the maximum value of  $\rho(t)$ , is approximately one order larger than that of  $\rho_{photo}$ . The simulated results indicate that the maximum value of the free-electron density  $\rho_{max}$  exceeds the critical value  $\rho_{crit}$  ( $10^{27} \text{ m}^{-3}$  at 1045 nm), and then decreases by the recombination. **Figure 8(b)** shows the maximum free-electron density plotted along the laser axis, showing  $\rho_{max}$  exceeds  $\rho_{crit}$  at locations  $z < 13.8 \mu\text{m}$ .

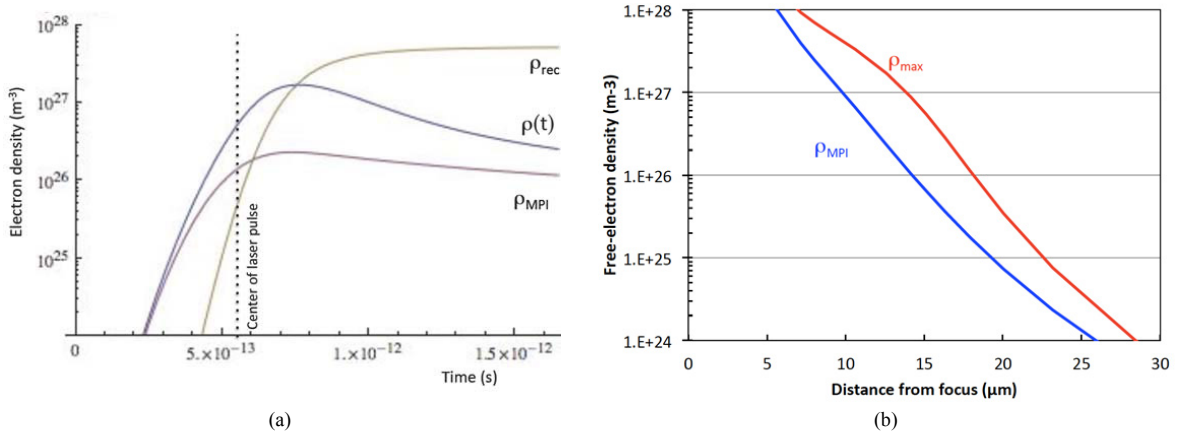


Fig. 8. Free-electron density with single laser pulse ( $Q = 2.56 \mu\text{J}$ ) simulated by rate equation given in Eq.(1). (a) Time-evolution of free electron density  $\rho(t)$  and  $\rho_{MPI}$  and  $\rho_{rec}$  at location of  $z = 12.6 \mu\text{m}$ ; (b) Maximum value of  $\rho_{max}$  and  $\rho_{MPI}$  plotted at different locations along laser axis assuming Gaussian beam propagation given in Eq.(4).

In reality, however, the free-electron density higher than  $\rho_{crit}$  is not available, because the plasma having critical free-electron density is highly reflective, and hence further increase in the free-electron density is prohibited. Then the laser energy reflected by the plasma region having  $\rho_{crit}$  is deposited in the upstream border of the plasma, and the free-electron density distribution is modified during laser pulse. Thus the plasma region with subcritical free-electron density moves toward the incoming laser source, leading to flattened free-electron density distribution. At the same time, the laser energy in the region underneath the subcritical plasma layer is blocked, and is hence cooled down by the thermal diffusion to the surrounding lower temperature region. Thus the lower boundary of the plasma also moves upward. It is concluded that the plasma moves upward as a whole, when critical free-electron density is reached in the plasma. The our above mode indicates that the moving velocity increases as the focus is approached, because more laser energy is reflected due to increased ionization rate, and accordingly cooling rate increases due to increased shielding effects. This agrees with the experimental result seen in region I in Fig. 4.

We should note, however, that the region with  $\rho_{photo} > 10^{26} \text{ m}^{-3}$  is limited only in a region of  $z > 13.8 \mu\text{m}$  in single pulse irradiation. This suggests that the upward motion per pulse caused by the flattening of the free-electron density is limited to the order of submicrons, and hence the upper edge of the plasma cannot reach  $z \approx 80 \mu\text{m}$  in single pulse irradiation seen in Figs. 2 and 3.

In order to produce critical free-electrons density in the plasma through avalanche ionization in a region of  $z > 13.8 \mu\text{m}$ , additional seed electron source other than MPI is needed. At high pulse repetition rate where heat is accumulated pulse by pulse, free-electrons can be also seeded by thermal excitation in the conduction band. The effect of thermally excited free-electrons on avalanche ionization has been discussed in several papers based on thermal diffusion model, where the inner structure is assumed to be filled completely by the plasma with the



temperature of  $\approx 3,600\text{ }^{\circ}\text{C}$  [9,22], which corresponds to the thermally excited free electron density of  $\rho_{\text{therm}} \approx 10^{25}\text{ m}^{-3}$  in D263 [9]. Our observation indicates that plasma temperature should be significantly higher than  $3,600\text{ }^{\circ}\text{C}$ , because the plasma is localized in the smaller volume, which is nearly 1 order smaller than the inner structure. Assuming free-electron density reaches  $\rho_{\text{crit}}$  through avalanche ionization with the increment of free-electrons density, it is expected the localized plasma is heated up to  $6,000\text{ }^{\circ}\text{C}$  by heat accumulation.

In Fig. 2, the plasma becomes brightest in a period II at #11~#13, which have some distance from the geometrical focus. The brightness of the plasma is determined by the competition between avalanche rate and the cooling rate between laser-pulses. While the former increases as the focus is approached, the latter shows the opposite tendency since the plasma volume increases due to increase in the spot size, which decreases the temperature gradient. As the result, the plasma becomes brightest at some distance from the focus as seen in Fig. 2, which agrees with intensity distribution of absorbed laser energy simulated based on thermal conduction model [9,22]. At height larger than the maximum brightness, the moving velocity of the lower edge becomes faster than moving velocity of upper edge, so that the upward motion becomes very slow (period III). When the plasma reaches some critical height where the laser irradiance is not strong enough to provide free-electron density of  $\rho_{\text{crit}}$ , the upward motion of the plasma is stopped. The decrease in the plasma temperature is accelerated because thermally excited free-electron density is decreased by the plasma cooling. Eventually the plasma by thermal diffusion ignites new plasma at the focus to start the new cycle.

#### 4. Concluding remarks

Dynamic behavior of plasma in internal modification of glass using *fs*-laser pulses at high pulse repetition rate up to  $1\text{ MHz}$  is characterized using high-speed video camera at 50,000 frames per second. The pictures reveal very striking fact that the plasma is locally produced in much smaller region than reported in literatures, and changes its location, size, shape and brightness periodically at repetition rates of the order of *kHz*, which is approximately three-orders smaller than laser pulses. The mechanism responsible for producing localized plasma and its periodic motion is analyzed qualitatively through the simulation of free-electron density based on a rate equation model, indicating that the upward motion occurs when supercritical density plasma is produced. The periodic motion of the plasma is produced because the laser-induced plasma moves toward incoming laser source up to the height where the plasma temperature is lowered to become transparent, when supercritical electron density of plasma is produced at high laser irradiances at high pulse repetition rates.

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